PROG6212_POE_PART 1

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Table of Content

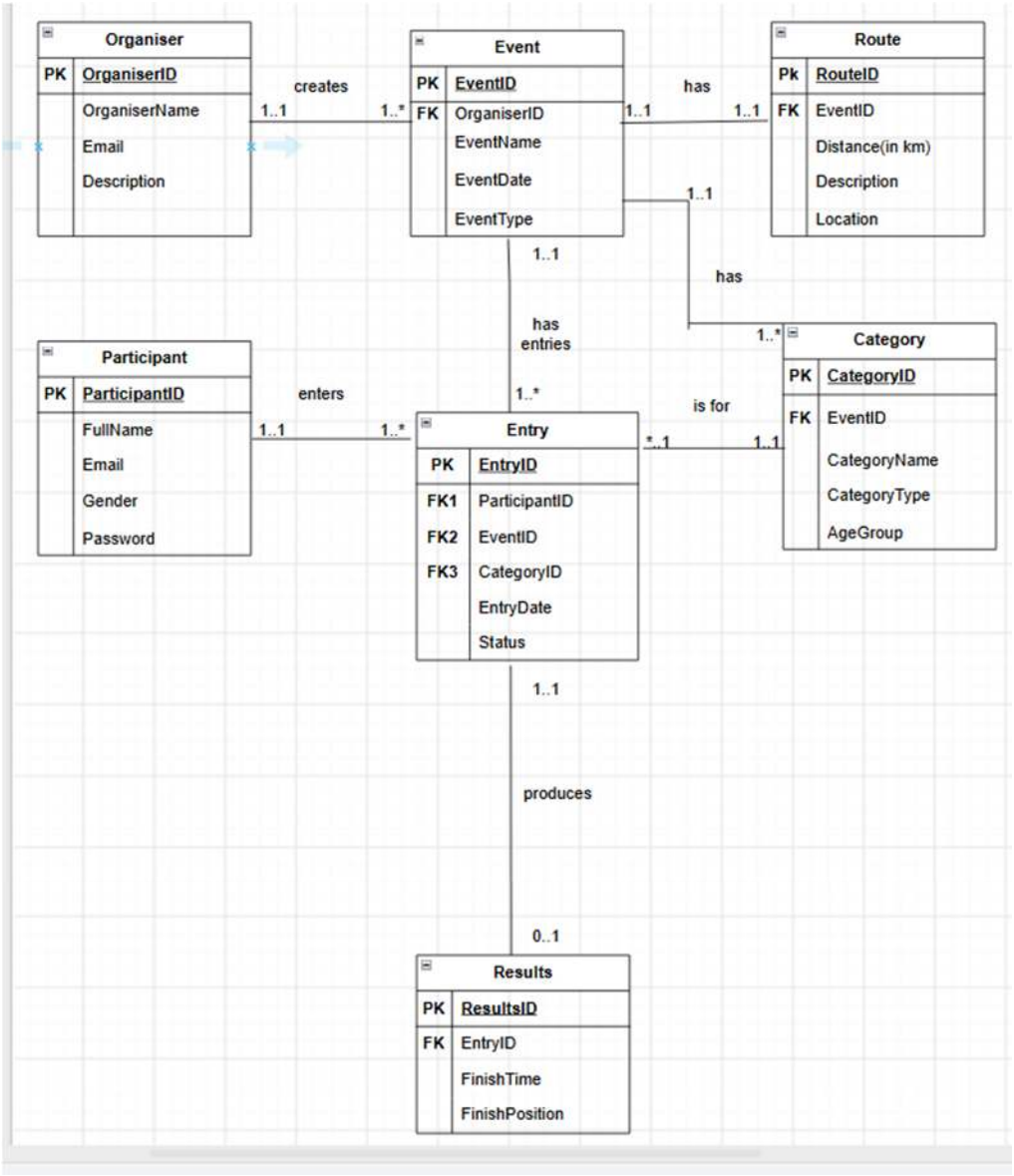
Section A
Entity Relationship Diagram.....3

Section B
Api Endpoint Plan.....4

Section C
SQL File Database.....8

Reference List 9

Section A - Entity Relationship Diagram



Section B_Api Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Ensures that a new user is registered	Public	FullName, Email, Password	201 Created_ User Account Approved
POST	/api/auth/logins	Gives registered users access to log in using their account credentials	Public	Email, Password	200 Ok_ Login Approved
GET	/api/profile	Users who have logged in are allowed to view their profile	Participant or Organizer	None	200 OK_ Displays the User Profile
PUT	/api/profile	Users who are logged in can update their profile details	Participant or Organizer	Profile details updated successfully	200 Ok_ Profile Update Confirmed
GET	/api/events	Returns all the events that are available	Organizer	None	200 Ok_ A list of events is displayed
GET	/api/events/{eventId}	Returns the details of a specific event	Participant or Organizer	None	200 Ok_ Displays the event that is selected

POST	/api/events	Creates a new event and presents the event details needed	Organizer	Name, description, date , location, distance , Event Type	201 Created_ The Event created is Approved
PUT	/api/events/{eventId}	Event details managed by the organizer are updated	Organizer	Event details are Updated	200 Ok_ Event Update Confirmed
DELETE	/api/events/{eventId}	Event details managed by the organizer are deleted	Organizer	None	204 No Content_ The Event is deleted successfully
GET	/api/events/{eventId}/categories	Returns categories that are available for the specific event	Participant/ Organizer	None	200 Ok_ Displays the event categories
POST	/api/events/{eventId}/categories	The distance or age category is created	Organizer	CategoryName, CategoryType, AgeGroup	201 created_ The Category is Created successfully
PUT	/api/categories/{categoiesId)	An existing category for the existing	Organizer	Category details are updated	200 Ok_ The Category Updated

		event requires updates			Successfully
DELETE	/api/categories/{categoiesId)	Category is removed from the event	Organizer	None	204 No content_ The Category is deleted successfully
POST	/api/events/{eventsId}/enrolment	A participant in an event is enrolled by selecting a category	Participant	CategoryId	201 Created_ Enrolment created successfully
GET	/api/events/{eventsId}/enrolment	All the participant enrolled in one of their events can be view by the organizer	Organizer	None	200 Ok_ The event enrolments are displayed
GET	/api/enrolments/{enrolment id}	The details of the specific enrolment are displayed	Participant/ Organizer	None	200 Ok_ Returns enrolment details
POST	/api/enrolments/{enrolment id}/results	The finishing time and finishing position are recorded	Organizer	FinishTime, FinishingPosition	201 Created_ Results Records are approved

		after an event			
GET	/api/results/profile	Gives the participant access to view their own event results	Participant	None	200 Ok_ Displays the participant results
GET	/api/events/{eventsId}/results	The Organizer now view the results recorded for their event	Organizer	None	200 Ok_ Displays the event results

Section C_ SQL File Database

```
--Creating Database--
Create Database RaceDayDB;
Use RaceDayDB;

--Creating Tables of the database--
Create Table Organiser
(
  OrganiserID INT Not Null,
  OrganiserName Varchar(50) Not Null,
  Description Varchar(255) Not Null,
  Email Varchar(100) Not Null UNIQUE,

  Constraint PK_Organiser
  Primary Key (OrganiserID)
);
Create Table Event
(
  EventID INT Not Null,
  OrganiserID INT Not Null,
  EventName Varchar(100) Not Null UNIQUE,
  EventDate Date Not Null,
  EventType Varchar(50) Not null,

  Constraint PK_Event
  Primary key (EventID),

  Constraint FK_Event_Organiser
  Foreign Key (OrganiserID)
  References Organiser(OrganiserID)
);
Create table Route
(
 RouteID INT Not Null,
```

```
  EventID INT Not Null,
  Distancekm Decimal(4,2) Not Null,
  Description Varchar(255) Not Null,
  location varchar(100) Not Null,

  Constraint PK_Route
  Primary Key (RouteID),

  Constraint Fk_Route_Event
  Foreign key (EventID)
  References Event(EventID)
);
Create Table Participant
(
  ParticipantID INT Not Null,
  FullName Varchar(100) Not Null,
  Email Varchar(100) Not Null Unique,
  Gender Varchar(10) Not Null,
  Password Varchar(100) Not Null,

  Constraint PK_Participant
  Primary Key (ParticipantID)

);
Use RaceDayDB;
Create Table Category
(
  CategoryID INT Not Null,
  EventID INT Not Null ,
  CategoryName Varchar(100) Not Null ,
  CategoryType Varchar(50) Not Null,
  AgeGroup Varchar(20) Not Null,

  Constraint PK_Category
  Primary Key(CategoryID),
```

```
  Constraint FK_Category_Event
  Foreign Key (EventID)
  References Event(EventID)
);

Create Table Entry
(
  EntryID INT Not Null,
  ParticipantID INT Not Null,
  EventID INT Not Null,
  CategoryID INT Not Null,
  EntryDate Date Not Null Default GetDate(),
  Status Varchar(20) Not Null Default 'Pending',

  Constraint PK_Entry
  Primary Key (EntryID),

  Constraint FK_Entry_Participant
  Foreign Key (ParticipantID)
  References Participant(ParticipantID),

  Constraint FK_Entry_Event
  Foreign Key (EventID)
  References Event(EventID),


  Constraint FK_Entry_Category
  Foreign Key (CategoryID)
  References Category(CategoryID)
);
Create Table Results
(
  ResultsID INT Not Null,
  EntryID INT Not Null UNIQUE,
```

```
  FinishTime TIME Not Null,
  FinishPosition INT Not Null,

  Constraint PK_Results
  Primary Key (ResultsID),

  Constraint Fk_Results_Entry
  Foreign Key (EntryID)
  References Entry(EntryID)
);
--Insert Function--
Insert Into Organiser
(OrganiserID, OrganiserName, Description, Email)
Values
(1, 'Njabulo Ndlovu', 'South African RaceDay Event',
'njabulo@ndlovu.co.za'),
(2, 'Lesego Leqola', 'Running Association',
'lesego@leqola.co.za');

Insert Into Event
(EventID, OrganiserID, EventName, EventDate, EventType)
Values
(1,1, 'City Run Of Johannesburg', '2026-09-27', 'Road Work'),
(2,1, 'Heritage Marathon' , '2026-10-13', 'Full Marathon'),
(3,2, 'Marathon Challenge', '2026-11-10','Road Running');

Insert Into Route
(RouteID, EventID, Distancekm,Description,location)
values
(1,1, 10.01, 'Maboneng to Constitution Hill','Johannesburg
CBD'),
(2,2, 15.04, 'FNB Stadium to Orlando Stadium','Soweto'),
(3,3, 21.13, 'South Gate to Mall of the South', 'Pretoria');

Insert Into Participant
```

```
(ParticipantID, FullName, Email, Gender, Password)
values
(1,'Zinhle Ngubane', 'z.ngubane@gmail.com',
'Female','Ngubane123'),
(2,'Prince Molapo', 'princemolapo2@gmail.com',
'Male','Molapo784');

Insert Into Category
(CategoryID, EventID, CategoryName,CategoryType, AgeGroup)
values
(1, 1, '20 km Open', 'RoadWork','18-35'),
(2 ,1 ,'15 km Masters', 'RoadWork','25-45'),
(3 ,2 ,'Half of Marathon is Open', 'RoadWork','40+'),
(4 ,2 ,'15 km Open', 'RoadWork','18-39'),
(5 ,3,'20 km Master', 'RoadWork','18-20'),
(6,3,'Half of  Masters', 'RoadWork','16-17');

Insert Into Entry
(EntryID, ParticipantID, EventID, CategoryID,EntryDate, Status)
values
(1,1,1,1, '2026-09-04', 'Approved'),
(2,2,1,1, '2026-09-04', 'Approved'),
(3,1,2,3, '2026-09-04', 'Approved'),
(4,2,2,3, '2026-09-04', 'Approved'),
(5,1,3,5, '2026-09-04', 'Approved');



Insert into Results
(ResultsID, EntryID, FinishTime, FinishPosition)
Values
(1,1, '23:52:25',12),
(2,2, '23:48:20',7);

--View Data in the tables--
Select * From Organiser;
```

```
Select * From Event;
Select * From Route;
Select * From Participant;
Select * From Category;
Select * From Entry;
Select * From Results;
```

Reference List

Draw.io Flowchart Maker Available at : https://www.bing.com/ck/a?!&&p=089cfced04fe45babb1f7a0dbf1e73901f3f861b6b749120ccfa0ba43c70f6d5JmltdHM9MTc4ODQ4MDAwMA&ptn=3&ver=2&hsh=4&fclid=3d74b98c-ad4b-6474-3b88-ae33aca165fe&psq=draw.io&u=a1aHR0cHM6Ly9hcHAuZGlhZ3JhbXMubmV0Lw (Accessed : 03 September 2026)

Lecture slides. Available at: WebAPI_ASPNETCore_PROG6212 - Programming 2B PROG6212 2026 FT DISD0601 RCGPBR Term2 GR03 (Accessed: 03 September 2026)